

RETAIL DEMAND INTELLIGENCE: FORECASTING & DEMAND DRIVER ANALYSIS

Inventory Planning, Pricing Dynamics
& Operational Performance



TOOLS USED

Python, Pandas, Numpy, Matplotlib, Plotly & ML libraries

PRESENTED BY

Tejas Jadhav

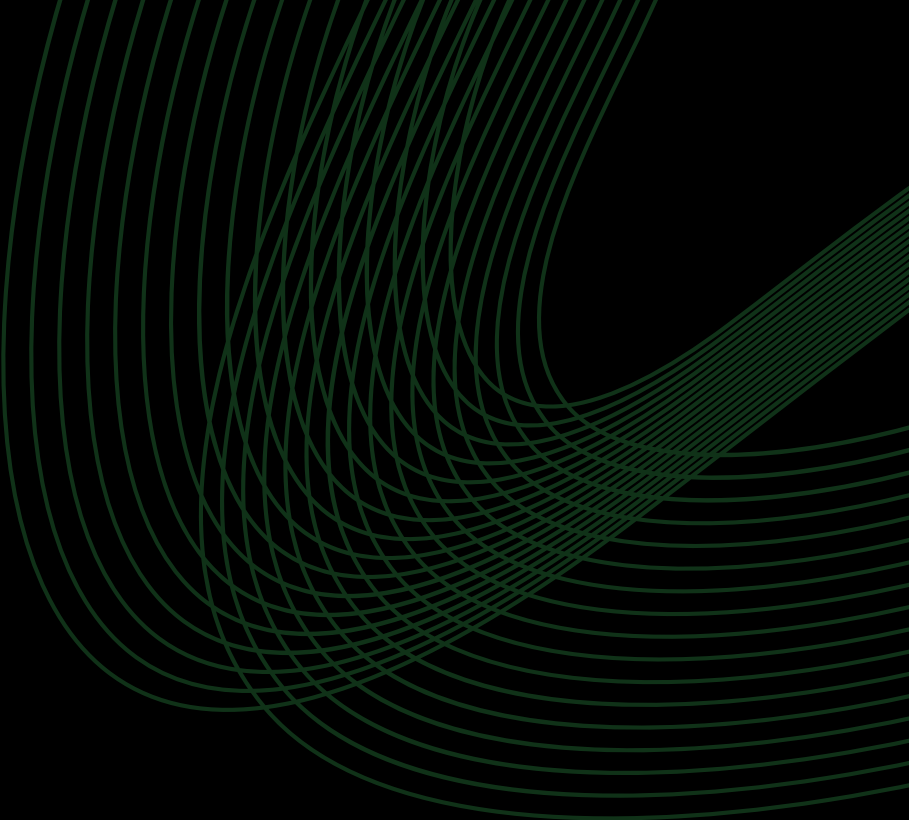
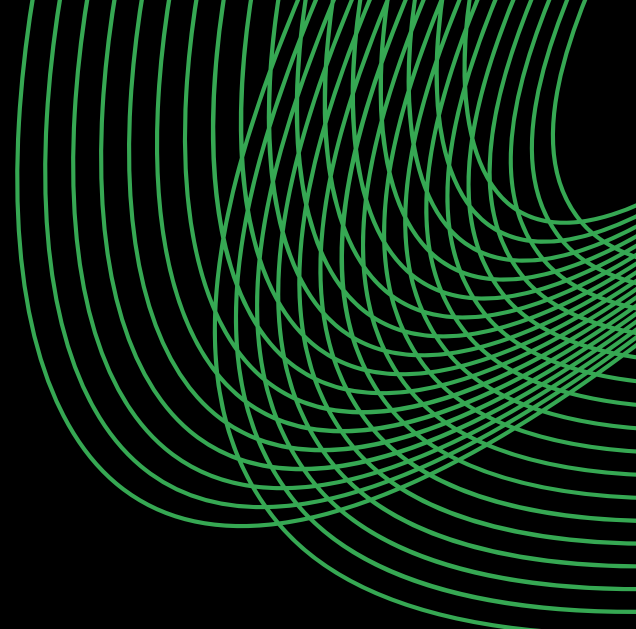


TABLE OF CONTENTS

Overview & Business Objective	3
Data Understanding & EDA	4
Data Preparation & Feature Engineering	5
Model Development & Experimentation	6
Model Evaluation & Comparison	12
Explainability & Final Model	34
Business Insights & Recommendations	36
Conclusion	37



Overview & Business Objective



1.1 PROJECT OVERVIEW

Retail businesses operate in a dynamic environment where demand is influenced by factors such as historical sales, inventory, pricing, promotions, seasonality, and region.

Accurate demand estimation helps businesses maintain optimal inventory, reduce stockouts and excess stock, and improve procurement and operational planning.

The Retail Demand Intelligence project uses historical retail data and machine learning to predict demand and identify the key factors influencing it.

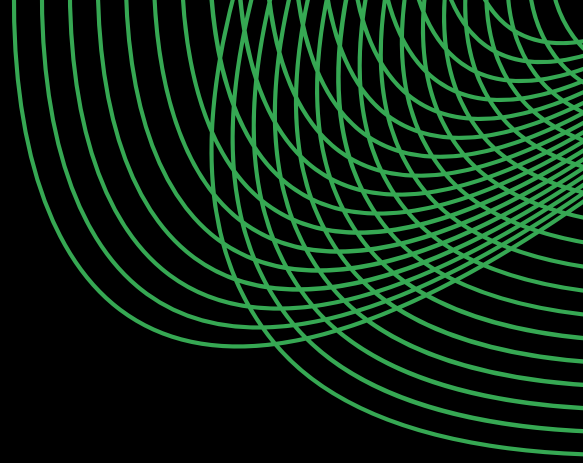
The project combines:

- Data understanding and cleaning
- Exploratory Data Analysis
- Feature engineering
- Machine learning regression
- Model evaluation and comparison
- Prediction-error analysis
- Model explainability
- Business-oriented insights

The objective is not only to predict demand accurately, but also to understand why the model makes its predictions & how these insights can support better decision-making.



1.2 PROBLEM STATEMENT



Retail demand is influenced by multiple interacting variables, making manual forecasting or reliance on historical averages insufficient for many situations.

The key problem addressed in this project is:

How can historical retail data be used to build a machine learning model that accurately predicts product demand while also identifying the major factors driving demand?

The project therefore treats Demand as the target variable and uses historical, operational, pricing, promotional, environmental, and temporal information as predictive features.



1.3 PROJECT OBJECTIVES

The major objectives of the project are:

- Develop a machine learning-based retail demand prediction system.
- Understand the underlying structure and characteristics of the dataset.
- Identify important patterns through exploratory data analysis.
- Create meaningful temporal, inventory, pricing, and historical features.
- Train and compare multiple regression models.
- Evaluate model performance using appropriate regression metrics.
- Investigate the largest prediction errors.
- Identify the most influential demand-driving features.
- Identify limitations and possible improvements for future development.



1.4 PROJECT SCOPE

The project follows a structured workflow covering data analysis, demand modeling, model explainability, and business interpretation to understand demand patterns, evaluate predictive performance, identify key demand drivers, and derive actionable retail insights.

01

Data Analysis

- Dataset inspection
- Data quality assessment
- Distribution analysis
- Relationship analysis

02

Demand Modeling

- Feature preparation
- Regression modeling
- Model comparison
- Model evaluation

03

Explainability

- Feature importance analysis
- Original-feature aggregation
- Investigation of large prediction errors

04

Business Interpretation

- Business Interpretation
- Inventory implications
- Pricing and promotion observations
- Seasonal and weather-related effects
- Recommendations for retail planning